INCH-POUND

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MILITARY SPECIFICATION

MARKING AND LABELING OF U.S. ARMY MARINE CRAFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the type, location, size, proportion, and color of markings (other than shipment markings) on Army Vessels.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1558	- Paint, Stencil.
H-B-621	- Brush, Stencil.
RR-S-714	- Stencil Set, Marking.
TT-E-489	- Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces).
TT-E-490	- Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings.
TT-T-291	- Thinner, Paint, Mineral Spirits, Regular and Odorless.
UU-S-625	- Stencilboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-514	- Plates, Identification, Instruction, and Marking, Blank.
MIL-A-2877	- Aluminum Alloy Tape.
MIL-D-16791	- Detergent, General Purpose (Liquid, Nonionic).
MIL-A-46050	- Adhesives, Cyanoacrylate, Rapid Room-Temperature curing, Solventless.
MIL-M-43719	- Marking Materials and Markers, Adhesive, Elastomeric, Pigmented; General Specification for.
MIL-M-43719/1	- Marking Materials and Markers, Adhesive, Elastomeric, Pigmented: Letters and Numerals.
MIL-M-43719/2	- Marking Materials and Markers, Adhesive, Elastomeric, Pigmented: Legends.
MIL-M-43719/3	- Marking Materials and Markers, Adhesive, Elastomeric, Pigmented: Symbols.
MIL-M-43719/4	- Marking Materials and Markers, Adhesive, Elastomeric, Pigmented: Hazardous Material Symbols.

STANDARDS

MIL-STD-12	- Abbreviations For Use On Drawings, and In Specifications, Standards and Technical Documents.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria For Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits For Army Materiel.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

MILITARY

U.S. Navy NAVSHIPS No. 810-1385848, Method of Marking Deck Plates for Sounding Pipes.

(Application for copies should be addressed to the Department of the Navy, Naval Ship Engineering Center, Washington, DC 20362.)

PUBLICATIONS

DEPARTMENT OF THE NAVY

General Specifications for Ships of the United States Navy 1977-1978 (NAVSEAS 0902-001-5000), Section 507, Section 602, and Section 664.

(Application for copies should be addressed to the Department of the Navy, Naval Ship Engineering Center, Washington, D.C. 20362.)

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DEPARTMENT OF TRANSPORTATION, COAST GUARD

Code of Federal Regulation 46 - Shipping.

(Application for copies should be addressed to the Office of the Federal Register, National Archives and Records Service, General Services Administration, Washington, D.C. 20408.)

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FMS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN BUREAU OF SHIPPING (ABS)

Rules for Building and Classing Steel Vessels

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. Marking and labeling of U.S. Army marine craft shall be as shown on the applicable figures and as specified herein.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The label plates shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the label plates may be exposed.

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3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the label plates may be newly fabricated from recovered materials to the maximum extent practicable, provided the label plates produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the label plates.

3.4 Application.

3.4.1 Letters, numerals, and other markings. Block lettering and numerals as shown in figure 1 shall be used for all exterior designation markings and draft numerals. Unless otherwise specified herein, all other lettering shall be Gothic capitals; all numerals shall be Arabic. Unless otherwise specified herein, colors shall black on a white or light-colored background, and white on a black or dark-colored background. Enamel paints conforming to TT-E-489 and TT-E-490 shall be considered a light background. All markings, label plates, and labels shall be located as specified in 3.4 through 3.11.5. If markings, label plates, or labels are obscured by the subsequent installation of curtains, rods, pipes, wiring, ducts, furniture, hull or deck fittings, canvas, or any other obstruction, they shall be relocated to a position as near as possible to the specified location and in such a manner and location that proper identification, maximum visibility, and legibility are ensured.

3.4.2 Labeling. Except where specified herein, all labeling shall conform to MIL-STD-1472, 5.5. Where noise levels exceed the values of MIL-STD-1474, 5.1.1, category D (85 dB(A)), noise hazard signs shall be as specified in MIL-STD-1474, 4.3.

3.4.3 Hull markings. Exterior markings of the hull shall consist of designation markings, draft numerals, and load-line markings and shall be painted the same color as the lettering. Outlines of markings on wood hulls shall consist of shallow scribe marks (maximum 0.125 inch (0.32 cm) wide by 0.125 inch (0.32 cm) deep). Outlines of markings on steel hulls shall consist of a fine bead weld in accordance with CFR 46, parts 41 to 69.

3.4.4 Pigmented elastomeric sheeting material, stenciling, and hand painting. Lifesaving and fire fighting equipment identification, locker and box content marking, nameboards, and signs in general, interior and exterior surfaces shall conform to MIL-M-43719, MIL-M-43719/1, MIL-M-43719/2, MIL-M-43719/3, and MIL-M-43719/4. Consideration shall be given to the condition of the surface upon which the identifying marking means is to be applied. Surface treatment shall conform to MIL-M-43719 for the pigmented, elastomeric, adhesive sheeting material and MIL-D-16791 for the painted identifying marking when the stenciling and hand

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painting method is employed. Paint for stenciled identifying marking shall conform to A-A-1558, class 1. The paint for hand painted letters shall conform to TT-E-489, class A. Stenciled identifying markings shall be in accordance with RR-S-714, H-B-621D, TT-T-291, and UU-S-625.

3.4.5 Label plates. Unless otherwise specified (see 6.2), label plates shall conform to the requirements specified in MIL-P-514. Label plates shall have corners rounded to a minimum of 0.03 inch (0.076 cm) radius. When label plates are mounted on metal they shall be secured by means of machine screws of a size commensurate with existing label plate hole size. When label plates are mounted on wood, round-head wood screws shall be used. Fastening means shall be electrochemically compatible with the label plates and the metal surface upon which the plate is mounted. When metal plates are to mounted where the screw holes can cause leaks — locations such as hulls, water tight or other liquid type bulkheads or decks — a metal plate of the required thickness for the label plate screw length with the required tapped holes shall be welded in place. A rapid room-temperature curing adhesive for metal label plates not exceeding 0.048 inch (0.12 cm) thick shall be used. Adhesive shall be in accordance with MIL-A-46050. Type I class designation shall be an option and governed by application. The adhesive is intended for attachment of metal label plates in place of mechanical attachment, suitable for exposure to severe use conditions, interior and exterior exposure, vibration, shock and high temperatures. Adhesive shall not be used for the following:

a. Propulsion plant identification.
b. Die-stamped label plates.
c. Indented corrosion resisting steel or nickel-copper label plates.
d. Flexible or curved surfaces.
e. Where the temperature can be over 200 °F (93 °C).

3.4.5.1 Photographic plate (photosensitive anodized aluminum). Photographic plates shall be used for dials, wiring and schematic diagrams, instruction plates, instrument panels, gauges, switch plates, and nameplates.

3.4.5.2 Color. Warning plates, when specified, shall have red lettering on a black background. All other plates shall have aluminum lettering on a black background.

3.4.6 Tags. Tags for electric cables and small-diameter pipes shall be aluminum tape conforming to MIL-A-2877. Lettering shall be 0.25 inch (0.635 cm) high, embossed at least 0.03125 inch (0.79 mil). Tags shall be as shown in figure 2.

3.4.7 Abbreviations. Abbreviations may be substituted for lengthy inscriptions, provided the meaning remains clear. Abbreviations shall be in accordance with MIL-STD-12 (see table II for system designations, 3.10.4.7).

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3.5 Designation markings. Markings shall be either black or white depending on the color of the hull. On a black hull markings shall be white. On a gray hull markings shall be black.

3.5.1 Self-propelled vessels.

3.5.1.1 Bow. Designations shall be as shown in figure 3 and shall be located on the hull or bulwark near the bow on both port and starboard sides. The legend shall be centered at three-fourths the height from the designed load waterline to the upper edge of the deck or bulwark. Where practical, the data shall be arranged in two lines. When the alternate one-line arrangement is used, the legend "U.S. ARMY" shall be to the left of the vessel name or number. Bow designations on tugs shall be located clear of permanently installed fender guards and bumpers.

3.5.1.2 Stern. Stern designation, "U.S. ARMY" and name or number shall be located on the hull or bulwark. On vessels having transom or elliptical sterns, designations shall be centered horizontally on the vertical centerline. On vessels having a cruiser stern, designations shall be located near the stern, on both port and starboard sides. Designations shall be centered at three-fourths the height from the designed load waterline to the upper edge of the deck or bulwark, whichever is higher. Size of the stern designation shall be as shown in figure 3.

3.5.1.3 Nameboards. When vessels 75 feet (23 m) or more in overall length have pilothouses, nameboards shall be provided. Only the name or number shall be put on the nameboard. Nameboards shall be located on the outboard side of the boat deck railing abreast of the pilothouse or on the outboard side of the pilothouse top railing. Nameboards shall be wood or steel and shall be rectangular in shape. The vertical dimension shall be twice the height of the lettering used. The designation marking shall be centered on the nameboard. The size of the lettering shall be as shown in figure 3, except that if the resulting nameboard would be disproportionately large, the lettering may be reduced in height to not less than one-half that specified. Nameboards shall have gold color lettering on a black background.

3.5.2 Nonpropelled vessels.

3.5.2.1 Ship-shaped hulls. Ship-shaped nonpropelled vessels shall have exterior designation markings as specified in 3.5.1.

3.5.2.2 Barge with long deckhouse. When length of the deckhouse is 40 percent or more of the overall length of the barge, designation marking shall be centered on the prot and starboard sides of the house. This same data shall also be centered on the forward and aft ends of the house. All markings shall be centered vertically at the midheight of the house or as nearly midheight as practicable. Sizes of designations shall be as shown in figure 3.

3.5.2.3 Barge with short deckhouse. When the length of the deckhouse is less than 40 percent of the overall length of the barge, designations shall be as specified in 3.5.2.4, except that the designation at the stern shall be centered on the aft end of the deckhouse.

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3.5.2.4 Barge without deckhouse. A barge without a deckhouse, deck cargo bins, or bulwarks shall have the designation marking centered longitudinally amidship on the port and starboard topsides. These data shall be centered vertically at 12 inches (30.48 cm) below the edge of the deck at side or 12 inches (30.48 cm) below the uppermost fender guard or rubbing strake. Size of designations shall be as shown in figure 3. Where practicable, these same data shall also be centered on the bow and stern immediately below the deck edge. If sufficient space is available on the bow and stern, markings shall be the same size as those used amidship. If sufficient space is not available, markings may be reduced in height to not less than one-half that specified for the amidship markings. Barge with deck cargo bins or bulwarks shall have the same designation markings, except that they shall be located at the midheight of the deck cargo bins or bulwark at the bow and stern and amidship, port and starboard.

3.5.2.5 Floating crane. The hulls of floating cranes shall have designations as specified in 3.5.2.4. In addition, the designations shall be centered on both sides of the crane cab. Size and form of designations shall be as shown in figure 3.

3.5.3 Sectionalized, knockdown, and collapsible craft. All sectionalized, knockdown, and collapsible type vessels shall have the applicable designation marking located as specified in 3.5.1 and 3.5.2, except where a specified legend on the side of the vessel would necessitate placing a part of the legend on each of two or more sections, the legend and number shall be placed so that it is entirely on one section.

3.6 Builder's nameplate.

3.6.1 Vessels and boats. Builder's nameplate for vessels and boats 40 feet (12 m) or more long shall be as shown in figure 4 except that the vessel's assigned name shall be included. Nameplates shall be of secondary cast brass, engraved nonferrous alloy plate of not less than No. 8 American Wire Gage (AWG), or embossed nonferrous alloy sheet of not less than No. 16 AWG. The nameplate shall be installed on the forward bulkhead of the engine casing or engine room, with nameplates installed at the extreme forward end of the vessel on the center deck plate.

3.6.2 Special vessels and small craft. Builder's nameplate for boats less than 40 feet (12 m) long, sectionalized, knockdown, and manually propelled craft, shall be as shown in figure 4 except that the vessel's assigned name shall be included. Each section of the sectionalized vessels and pontoons shall have a nameplate attached to the inner side of a transverse or longitudinal bulkhead just below the main deck level and near the main access opening to space below decks in vessels having decks, or at the centerline of the unit in boats or pontoons without decks.

3.7 Load line and draft markings. Markings shall be either black or white depending on the color of the hull. On a black hull markings shall be white. On a gray hull markings shall be black.

3.7.1 Load line markings. Large self-propelled barges, tugs, and similar vessels shall have load line markings as specified in Rules for Building and Classing Steel Vessels.

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3.7.2 Draft numerals. Draft numerals are mandatory for all vessels, boats, or barges with a load waterline length of more than 45 feet (13.72 m). Numerals shall be Arabic, and shall be 6 inches (15.24 cm) high, spaced at 1-foot (30.48-cm) intervals vertically.

3.7.2.1 Vertical location. Draft numerals shall be located so that the bottom edge of each numeral will establish the draft indicated by that numeral. The bottom edge of the numeral shall be measured in exact feet, normal to the designed load waterline, and above the forward and aft reference points. Numerals shall extend from one numeral below the lightest draft to one numeral above the deepest draft at which the vessel may float at bow or stern.

3.7.2.2 Forward draft numerals. The forward draft numerals shall be placed on port and starboard as near the stem as practical. The vertical draft marks shall be located in a vertical line which passes through a point established by the intersection of the design load waterline and the forward edge of the stem. The reference and starting point from which the vertical measurements for the forward draft marks will be determined shall be the intersection of the forward vertical line and the horizontal extension line from the point where the stem makes an abrupt upward curvature from the rear horizontal lower line of the keel. For a straight horizontal keel, this line is an extension of the keel bottom.

3.7.2.3 After draft numerals. The after draft numerals shall be placed port and starboard as near the aft vertical as practicable. On vessels with transom sterns, the numerals shall be located horizontally on the transom at the vertical centerline of the vessel. The reference point from which the vertical measurements for the after draft marks are determined shall be the lowest point of the keel, propeller guard, or stern frame. On those vessels where the propeller extends below the lowest extremity of the hull or appendages, the reference point shall be the lowest point of the propeller circle.

3.8 Lifesaving and fire fighting equipment.

3.8.1 Lifeboats. Hull markings for lifeboats and other boats carried on board shall consist of "U.S. ARMY" and the number of the parent vessel, with letters 2 inches (5.08 cm) high. The markings shall be both forward and aft; if double "bowed", on the prot and starboard. On transom-sterned boats, mark at the bow and at the center of the transom. The numbers of each lifeboat shall be stenciled or marked on both sides of the bow in numerals 3 inches (7.62 cm) high. The cubic contents and the maximum number of persons allowed to be carried in each lifeboat shall be marked or stenciled on each side of the bow in letters and numbers 1.5 inches (3.81 cm) high. In addition, the maximum number of persons allowed shall be plainly marked or stenciled on top of two or more of the thwarts, in letters and numbers 3 inches (7.62 cm) high. To ensure that the external markings shall not be obscured when the lifeboat covers are installed, lifeboat markings shall begin 8 inches (20.32 cm) or more below the gunwale. The lifeboat oars shall be stenciled with the words "U.S. ARMY" followed by the name or number of the parent vessel in letters and numbers 1 inch (2.54 cm) high. The markings shall be black.

3.8.2 Other lifesaving equipment. Markings for life floats and life rafts carried on board shall consist of "U.S. ARMY" and the number of the parent vessel clearly stenciled on them in letters 2 inches (5.08 cm) high. Stenciling shall appear in the center of the front and rear. The maximum number of persons

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allowed shall be stenciled on the top of the sides in letters 3 inches high (7.62 cm). The oars shall be stenciled as specified in 3.8.1. The markings shall be black on floats and rafts.

3.8.3 Fire fighting equipment. Fire hydrants, fire valves, and chemical fire fighting control station equipment shall have the words "FIRE HYDRANT" or "FIRE VALVE," or other applicable designation, and the applicable fire station number in 2-inch (5.08 cm) red letters placed immediately above the fire fighting equipment on the bulkhead on which they are mounted. Hose racks shall be marked "FOR FIRE" in 1-inch (2.54-cm) letters. Valves shall be provided with double-ended arrows showing the direction of operation and labeled at each end to indicate the functional result (e.g. open, close, on, off, etc.).

3.8.4 General alarm bells. General alarm bells shall have "GENERAL ALARM BELL - WHEN BELL RINGS GO TO YOUR STATION" painted in 1-inch (2.54-cm) red letters beneath the bells on the bulkhead on which they are mounted.

3.8.5 Engine room and painting compartments. Vessels having engine rooms, painting compartments, or other space where personnel work and which are equipped with fixed CO_2 fire fighting systems shall display the following warning statement using signs or bulkhead marking: "WARNING - WHEN SIREN SOUNDS, VACATE AT ONCE. CARBON DIOXIDE BEING RELEASED." The red lettering shall be 2-inch (5.08-cm) high minimum and shall be conspicuously located.

3.8.6 Crash panels. Crash panels and emergency escapes shall have "EMERGENCY EXIT" painted on them in 2-inch (5.08-cm) letters in locations as follows:

a. On compartment side of fire screen doors.
b. On corridor side of stairwell doors.
c. On inside of stairwell doors leading to boat deck.

Color shall be that which will be most legible against the color of the respective doors and degree of light; however, red will be used where possible.

3.8.7 High-voltage hazard spaces. Entrances to electrical controller rooms, high-voltage switchboards, and other high-voltage spaces shall display the following warning in letters not less than 1 inch (2.54 cm) high: "DANGER - HIGH VOLTAGE - AUTHORIZED PERSONNEL ONLY." Such warnings shall be painted on the doors and access, or on a adjacent bulkhead, whichever is more conspicuous, using a contrasting color; however, red will be used where possible.

3.9 Compartmentation labeling.

3.9.1 Bulkhead and frame numbers. All vessels, except sectionalized barges and boats less than 30 feet (9.14 m) overall length, shall have bulkhead and frame numbering plates. Each complete bulkhead shall be numbered, and where complete bulkheads are more than 20 feet (6.10 m) apart, partial bulkheads, joiner bulkheads located on frames, or frames located in intermediate positions shall be labeled so that bulkhead or frame numbers are not more than 20 feet (6.10 m) apart. Bulkhead and frame numbers shall correspond to numbers shown on "as constructed" drawings and shall be written on one line, using abbreviations "BHD 12," "FR 12," or other applicable abbreviations. Bulkhead numbers shall be

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located as conspicuously as possible, over important passageways or at the centerline of the ship, at or above eye level. Frame numbers shall be located on deck beams as near to the centerline of the ship as possible. All bulkhead and frame number plates shall be fastened to the midship side of the bulkhead or frame. Dimensions of label plates and lettering shall be as shown in figure 5.

3.9.2 Door, manhole, hatch, and tank labeling. Doors, manholes, and hatches shall be labeled to indicate the compartment being entered. If a door, manhole, or hatch opens into a passageway or trunk to which several compartments are connected, the principal compartment shall be included on the label. For tanks, the plate shall also include the capacity in gallons.

3.10 Piping system.

3.10.1 Pipelines. Pipeline marking and flow indicating will be included on all pipelines. The selection of the color and system designation shall be in accordance with the General Specification for Ships of the United States Navy, Section 507. All paint color numbers shall be in accordance with TT-E-489.

3.10.2 Color coding. Color coding of pipe shall be in accordance with table I.

TABLE I. Color coding of pipeline.

Pipeline fluid	Color	Paint number
Potable fresh water	Light blue	15200
Non-potable fresh water	Dark blue	15044
Sea water	Dark green	14062
Steam & Steam drains	Black	17038
Ship service air	Tan	10324
High pressure air	Dark gray	16081
Gasoline	Yellow	13538
Fuel oil	Yellow	13538
Lube oil	Yellow	13538
Hydraulic fluid	Orange	12246
Freon	Dark purple	17100
Sewage	Gold	17043
Waste drain water	Light gray	16251
Halon	Striped, gray/white	16187/17886
Fire lines & plugs	Red	11105

3.10.3 Pipe marking. In addition to special colors required for particular systems, all systems shall be marked for identification. This marking shall be the functional name of the system; where necessary for differentiation between two or more pipes of the same system, the specific services shall be included. As an example, plumbing drains shall be marked "soil drain" or "waste drain" to suit the type of drainage carried in the pipe. In addition to the specific service, the system's pressure shall be indicated: for example "600 lb/in² steam to forced draft blowers." For an outside diameter of 2 inches (5.08 cm) and larger (bare or lagged) pipe markings shall be painted on pipes or lagging using

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stenciled letters 1 inch (2.54 cm) high. For smaller pipes the lettering size shall be reduced, as necessary, but shall not be less than 0.375 inch (0.85 cm) high. Direction-of-flow arrows shall also be painted on. Where pipe sizes are too small for lettering, label plates shall be wired in place.

3.10.3.1 Method of marking. The marking shall be applied with paint and use of stencils, or vinyl cloth markers may be used. Marking shall be white for oxygen piping and black for other systems, unless the painted color of the pipe is so dark that white must be used for the marking to be legible.

3.10.3.2 Marking of pipe in unmanned spaces. Marking of piping located in tanks, voids, cofferdams, bilges and similar unmanned spaces shall be by label plate curved to fit the piping and wired in place on the piping. The label plate and wire shall be made of materials which are compatible to both the pipe and the fluid contained in the space. Marking shall be made once in each space. For machinery spaces, marking shall be made at entry and at exit. Where the service of the pipe is obvious, such as a short drain from a tank pipe, marking is not required. Where piping is behind protective battens, one label plate shall be attached to the batten listing all pipes behind the batten. Where marking of pipes on weather decks is required, it shall be by label plate.

3.10.3.3 Method of marking deck plates. The method for marking deck plates for sounding pipes shall be as shown on drawing NAVSHIPS No. 810-1385848.

3.10.4 Valve marking. All automatically operated valves, such as relief, pressure-regulating, pressure reducing, and back-pressure valves, shall have identification plates. Manually operable valves in piping shall have a label plate describing the functional service of the valve and shall be provided with a double-ended arrow showing direction of operation and labeled at each end to indicate the functional result (e.g. open, close, on, off, etc.). Where a valve is installed below a floor plate or grating, or is similarly hidden, a marking shall be installed on both the valve and the portable access plate. Individual valves in manifolds shall be separately marked, and an information plate indicating the valve combinations for specific services shall be installed wherever necessary.

3.10.4.1 Marking of valves in systems extending outside machinery spaces. Marking of all valves in systems extending outside machinery spaces (such as fire main, main drainage, fuel transfer, compressed air) other than those associated with the propulsion plant shall comply with the following:

The system name, the functional service, where appropriate the name and number of the tank, compartment, gun, or component served, and the basic location number of the valve shall be indicated for each valve. This includes valves associated with damage control.

3.10.4.2 Damage control valves. Damage control valves are those for which damage control diagrams are required to be prepared in accordance with section 664 of General Specifications for Ships of the United States Navy. See also section 602 for damage control closure classification labels.

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3.10.4.3 Valve numbering. Valves in each system shall be numbered without respect to valves in other systems: That is, there may be several valves carrying that same number, but they will be differentiated by the system designation of the valve and piping.

3.10.4.4 Fire plug numbering. Fire plugs require no system name. However, in addition to their functional service, fire plugs and their basic location number, the designation (name and basic location number) of the cutout valve serving the fire plug shall be indicated.

3.10.4.5 Remote valve operation numbering. Where both local and remote valve operation is installed, the basic location number at both places of operation shall be that of the valve.

3.10.4.6 Unmanned space valve numbering. Where valves are located in tanks, voids, or other normally unmanned spaces, the basic location number indicated shall be that of the place of remote operation.

3.10.4.7 System designation propulsion plant valves. System designation propulsion plant valves shall be in accordance with table II.

TABLE II. System designations.

System	Designation
Auxiliary machinery sea-water cooling	ASW
Control air	CA
Diesel air starting	DA
Diesel fresh water cooling	DFW
Diesel lube-oil service and transfer	DLO
Diesel oil service and purification	DOS
Diesel sea-water cooling	DSW
Fresh water drain collecting	FWDC
Fuel service	FOS
Gland seal and leak off	GS
Lubricating oil purification	LO
Main lubricating-oil service	LOS
Main machinery sea-water cooling	MSW
Shaft lubrication-oil service	SLO

3.10.4.8 Propulsion plant valves. Since the valves for the piping systems are associated with propulsion plant systems, they shall not have basic location numbers assigned. These valves shall be identified by a three-part designation in the following sequence:

Shaft or plant number;
System designation letters;
Individual valve number.

Individual valve numbers shall be assigned in sequence from the origin of a system to its terminus, exclusive of branch lines.

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3.10.4.9 Method of marking valves. The method of marking valves shall be by marking inscribed on rims of handwheels, or by marking on a circular label plate attached by the securing nut on the handwheel. If size or location prevents the sue of one of these methods, marking inscribed on a rectangular label plate to fit the pipe adjacent to the valve, wired in place for easy reading, shall be used. An insulating material shall first be wrapped around the bare pipe to prevent electrolytic action between pipe, label, and wire. When used, wire shall not cause electrolytic action between the label, pipe, and wire.

3.10.4.10 Label inscriptions. The inscription shall indicate specifically the designation of the fitting. For example:

"Fire Main Cutoff"
"Potable Water"
"Non-Potable Water—Unfit for Drinking"
"Soil Drain"
"Waste Drain"
"600 lb/in² Steam"
"High Pressure Air"
"High Pressure Hydraulic Oil"
"Oxygen"
"Halon"
"Sea Water"

3.10.4.11 Method of marking floodcocks, sea valves, deckplates and air escapes at fuel oil tanks. Floodcock and sea valve labels, and labels on valves, deck-plates, and air escapes at fuel oil tanks shall have depressed lettering, and the depressions shall be filled in with red enamel.

3.10.4.12 Special deckplate inscriptions. Deckplates over sounding tubes, tank fill pipes, and valve operating gear shall be inscribed with the name of the sounding tube, the tank number and contents, or operating valve reached through the deckplates in 0.25 inch (0.635 cm) letters and in abbreviations if practicable.

3.10.4.13 Manifold labels. Manifold labels shall be located on the bulkhead over the center of the group of valves, and just above the valve label plates. If there is no adjoining bulkhead, manifold label plates shall be located on the manifold box or box casing. The plates shall be as shown in figure 6. In addition to the manifold label, each valve in the manifold shall be labeled. For valves with wheels located beneath the floor plates, a label plate repeating the label on the valve shall be set into the top of the floor plates close to the cover giving access to the valve.

3.11 Miscellaneous labeling.

3.11.1 Electrical cable tags. Electrical cable tags as specified in 3.4.6 shall be installed on all cables, except the non-vital ones which are located wholly within a compartment in such a manner that they can be traced. Cables shall be tagged as near as practicable to each point of connection, on both sides of decks, bulkheads, and other barriers, and at intervals such that the length of cable between cable tags does not exceed 50 feet (15.24 m).

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3.11.2 Hoist load limits. Each davit, boom, hoist, derrick, and hoisting mechanism shall be provided with a label plate. The plate shall indicate "Safe Working Load," followed by the number of pounds or long tons maximum load the item is designed to carry. Lettering shall be 0.5 inch (1.27 cm) high.

3.11.3 Box and locker markings. The contents of each deck box, locker, and storage container for general use shall be clearly identified. Marking shall be stenciled letters 3 inches (7.62 cm) high, specifying contents of box or locker and stowage capacity in pounds, bushels, or number of units stowed. Marking shall be located in center of top or front, and on sides or ends, as the size and visibility of the box or locker indicates.

3.11.4 Warning boards. Tank ships, oil barges, and similar vessels which carry flammable, explosive, or other dangerous materials shall be equipped with portable boards made of exterior plywood or sheet steel. The boards shall clearly state:

WARNING
NO OPEN LIGHTS
NO SMOKING
NO VISITORS

Letters shall be a minimum of 2 inches (5.08 cm) high, except that the word "WARNING" shall be a minimum of twice the size of the other lettering.

3.11.5 Special labeling. Any special markings required (see 6.2) for ease of maintenance or repair shall be applied in a manner consistent with similar type markings.

3.12 Workmanship. The fabrication and installation of label plates and the application of labeling, marking, and identification shall be consistent with standard marine practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an

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acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article unit shall be examined as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Marking and labeling of each marine craft shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

MAJOR

101. Materials not as specified (see 3.3).
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
104. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.1.2).
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the label plates (see 3.3.2).
106. Letters and numbers not as shown on applicable figures (see 3.4.1).
107. Colors of background not as specified (see 3.4.1).
108. Colors of letters and numerals not as specified (see 3.4.1).
109. Markings, label plates, and labels not located as specified (see 3.4.1).
110. Labeling not as specified (see 3.4.2).
111. Exterior designating markings, draft numerals, and load markings not applied as specified (see 3.4.3).
112. Outlines of exterior hull markings not scribed, welded, or painted as specified (see 3.4.3).
113. Stenciling and hand painting not as specified (see 3.4.4).
114. Label plates made of material other than that specified (see 3.4.5).
115. Label plates not secured as specified (see 3.4.5).
116. Phenolic plates used in locations exposed to weather (see 3.4.5).
117. Photographic plates not as specified (see 3.4.5.1).
118. Warning plate color not as specified (see 3.4.5.2).
119. Tags not as specified (see 3.4.6).
120. Unauthorized abbreviations used (see 3.4.7).

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121. Designation marking color not as specified (see 3.5).
122. Bow designation not as specified (see 3.5.1.1).
123. Stern designation not as specified (see 3.5.1.2).
124. Nameboards not as specified (see 3.5.1.3)
125. Nonpropelled ship-shaped hull not designated as specified (see 3.5.2.1).
126. Nonpropelled barge with long deckhouse not designated as specified (see 3.3.2.2).
127. Nonpropelled barge with short deckhouse not designated as specified (see 3.5.2.3).
128. Nonpropelled barge without deckhouse not designated as specified (see 3.5.2.4).
129. Floating crane not designated as specified (see 3.5.2.5).
130. Sectionalized, knockdown, and collapsible craft not designated as specified (see 3.5.3).
131. Builder's nameplate on a vessel or boat, 40 feet (12 m) or more in length, not as specified (see 3.6.1).
132. Builder's nameplate on a special vessel or small craft, less than 40 feet (12 m) in length, not as specified (see 3.6.2).
133. Color of load line and draft markings not as specified (see 3.7).
134. Load line markings not as specified (see 3.7.1).
135. Draft numerals not as specified (see 3.7.2).
136. Vertical location of draft numerals not as specified (see 3.7.2.1).
137. Forward draft numerals not as specified (see 3.7.2.2).
138. After draft numerals not as specified (see 3.7.2.3).
139. Lifeboats not marked as specified (see 3.8.1).
140. Other lifesaving equipment not marked as specified (see 3.8.2).
141. Fire fighting equipment not marked as specified (see 3.8.3).
142. General alarm bells not marked as specified (see 3.8.4).
143. Engine room and painting compartment not marked as specified (see 3.8.5).
144. Crash panels not marked as specified (see 3.8.6).
145. High-voltage hazard spaces not marked as specified (see 3.8.7).
146. Bulkhead and frame plates not marked as specified (see 3.9.1).
147. Doors, manholes, hatches, and tanks not labeled as specified (see 3.9.2).
148. Pipelines not marked as specified (see 3.10.1).
149. Pipe color coding not as specified (see 3.10.2).
150. Pipe differentiation not as specified (see 3.10.3).
151. Method of marking pipes not as specified (see 3.10.3.1).
152. Marking of pipes in tanks, voids, cofferdams, bilges, and unmanned spaces not as specified (see 3.10.3.2).
153. Method of marking deck plates not as specified (see 3.10.3.3).
154. Valve marking not as specified (see 3.10.4).
155. Marking of valves outside of machinery spaces not as specified (see 3.10.4.1).
156. Damage control valves not as specified (see 3.10.4.2).
157. Valve numbering not as specified (see 3.10.4.3).
158. Fire plug numbering not as specified (see 3.10.4.4).
159. Remote valve operation numbering not as specified (see 3.10.4.5).
160. Unmanned space valve numbering not as specified (see 3.10.4.6).
161. Propulsion plant valves not designated as specified (see 3.10.4.7).
162. Propulsion plant valves not identified as specified (see 3.10.4.8).
163. Method of marking valves not as specified (see 3.10.4.9).
164. Label inscriptions not as specified (see 3.10.4.10).
165. Method marking floodcocks, sea valves, deckplates, and air escapes not as specified (see 3.10.4.11).

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166. Special deckplate inscriptions not as specified (see 3.10.4.12).
167. Manifold labels not as specified (see 3.10.4.13).
168. Electrical cable tags not as specified (see 3.11.1).
169. Hoisting mechanisms not marked as specified (see 3.11.2).
170. Boxes and lockers not marked as specified (see 3.11.3).
171. Warning boards not marked as specified (see 3.11.4).
172. Special markings not as specified (see 3.11.5).
173. Workmanship not as specified (see 3.12).

5. PACKAGING

Not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification is intended for use in the marking and labeling of U.S. Army marine vessels. It is not applicable to vessels of other activities or services.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Timeframe required for submission of first article label and number of labels required (see 3.2).
d. Label plates, if other than in accordance with MIL-P-514 (see 3.4.4).
e. Special marking required (see 3.11.5).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction unit. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Identification, ship
Marking, designation
Marking, hull
Marking, marine
Marking, propulsion plant
Nameboards
Plate, label
Stenciling

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 AQL certification. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQL's/LTPD's (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications.
(Certified by/date: _____). 7/14/89

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
DLA - CS

Project 1910-0002

User activity:
Navy - OS

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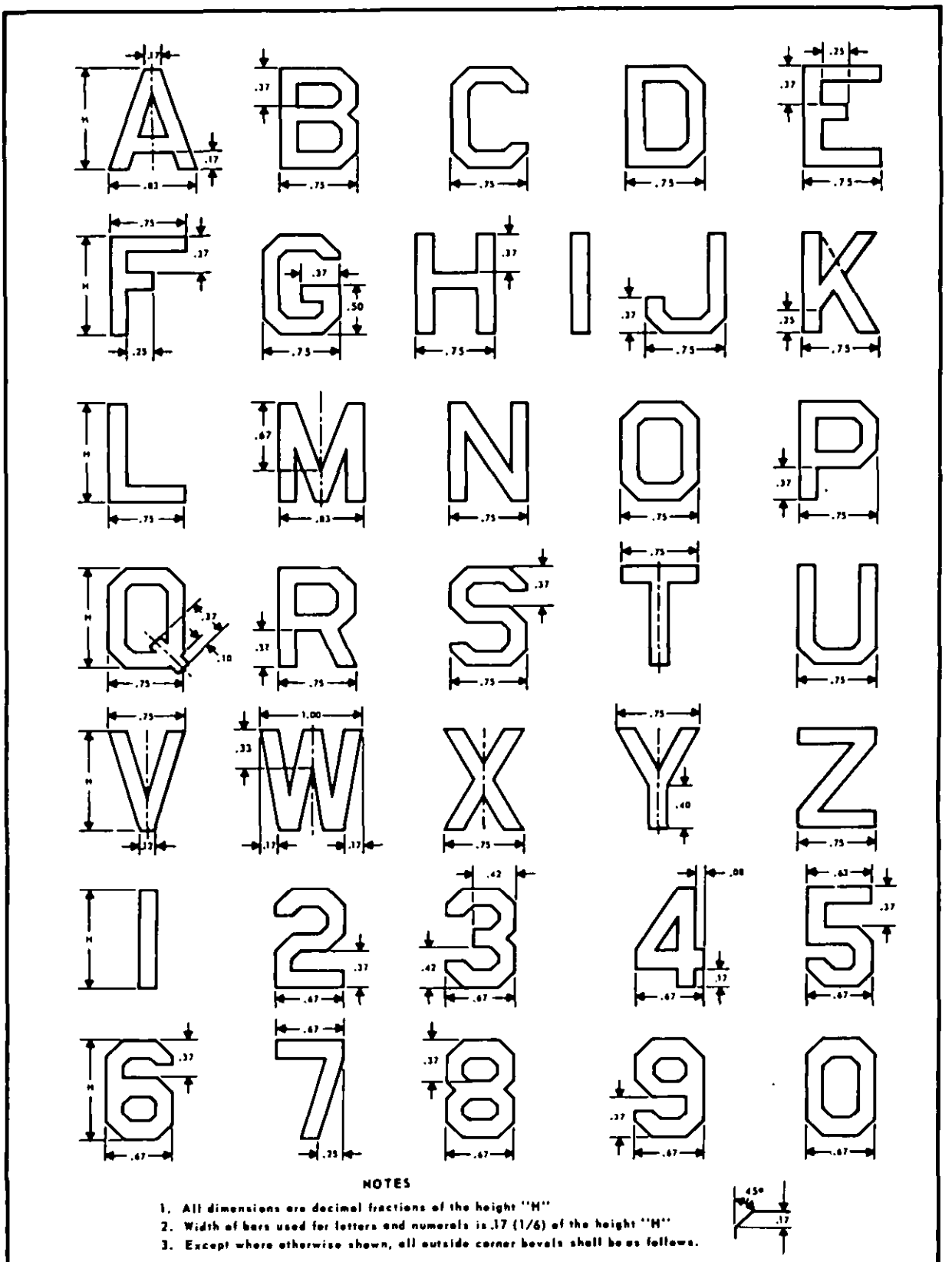
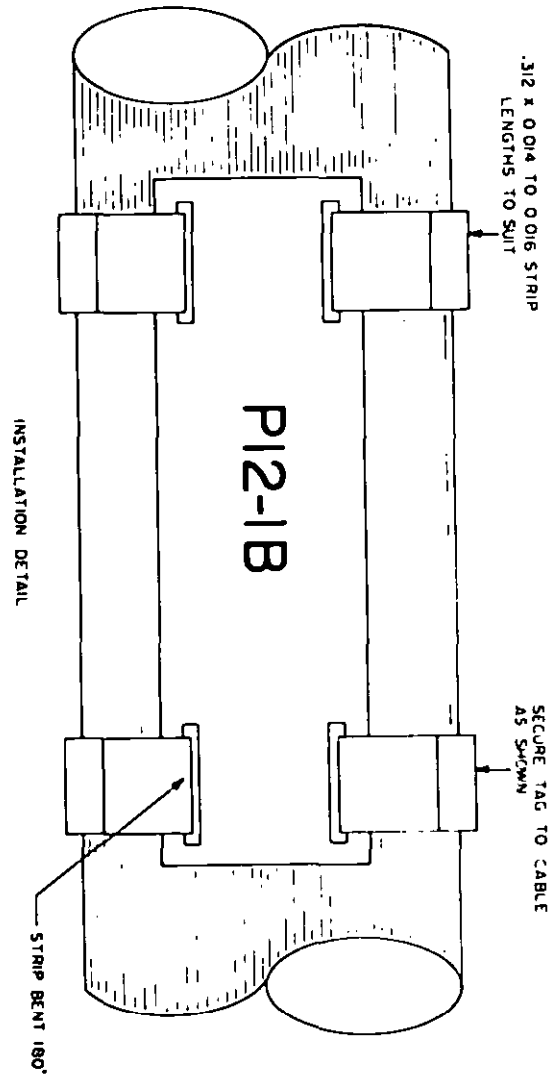
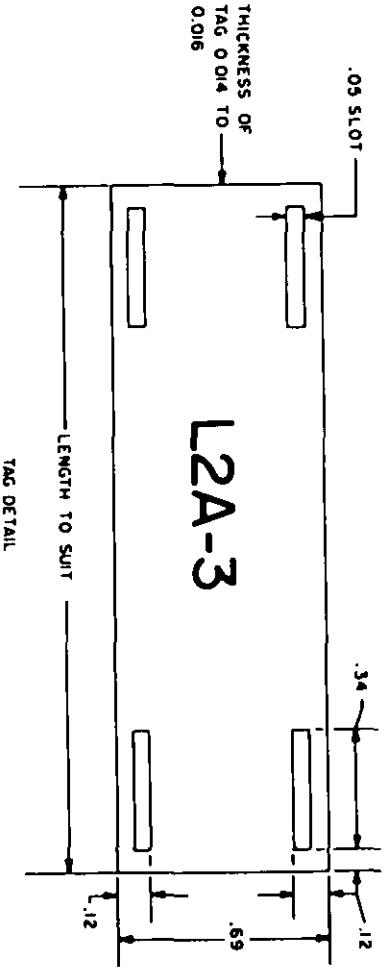


FIGURE 1. Block letters and numerals.

X-836

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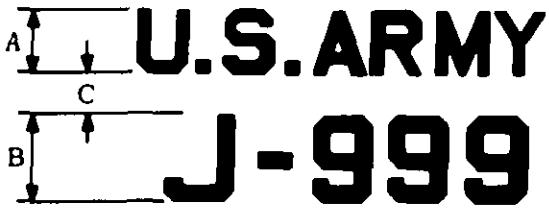


NOTE:
ALL DIMENSIONS ARE IN INCHES.

FIG.2 Typical cable tags.

X-837 C

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PREFERRED DESIGNATION ARRANGEMENT



ALTERNATE DESIGNATION ARRANGEMENT

TABLE 1. DIMENSIONS FOR DESIGNATIONS

SIZE OF VESSEL BOAT OR BARGE LOAD WATER LINE LENGTH	DIMENSIONS IN INCHES		
	A	B	C
LESS THAN 20 FEET	3	4	2
20 FEET TO LESS THAN 40 FEET	4.50	6	3
40 FEET TO LESS THAN 70 FEET	6	8	4
70 FEET TO LESS THAN 125 FEET	7.50	10	5
125 FEET AND OVER	9	12	6

FIGURE 3. Designation arrangements.

X-838 A

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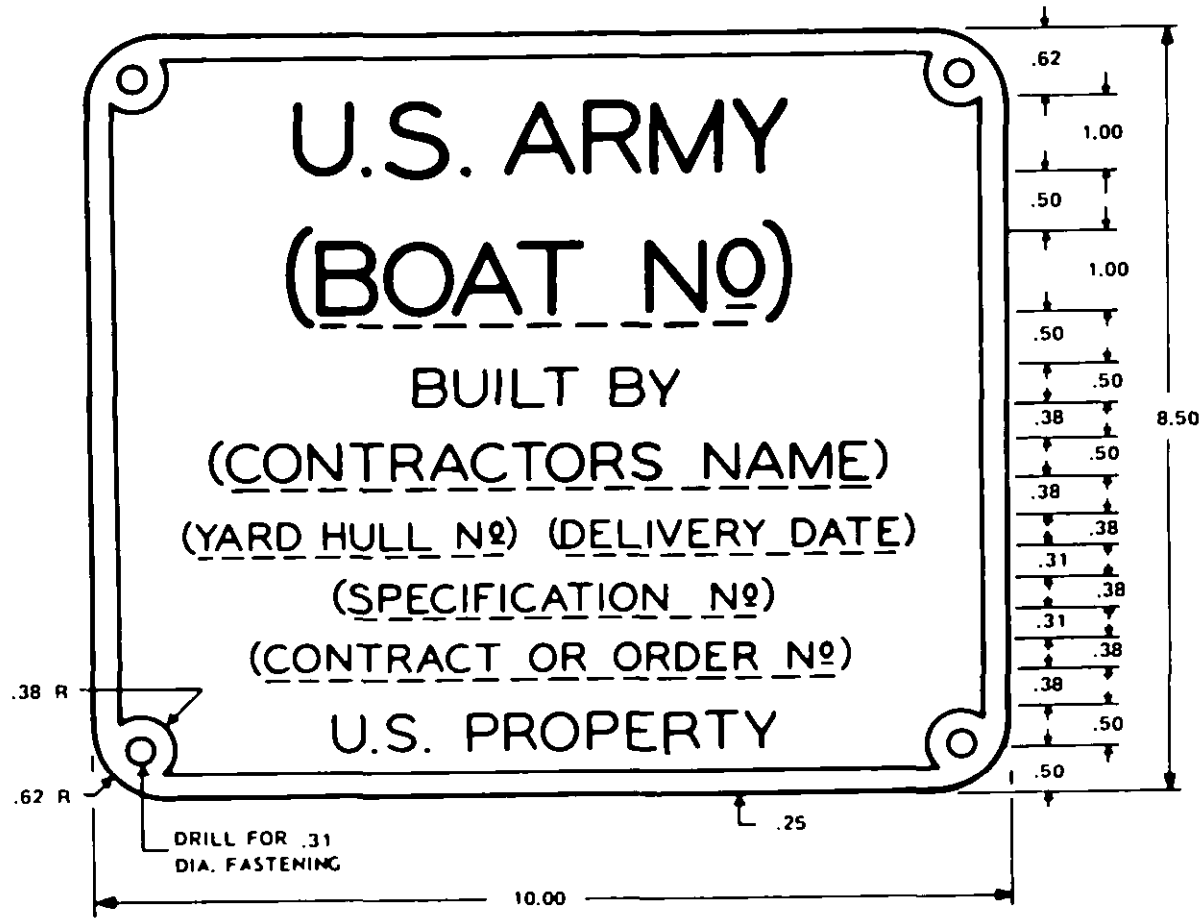
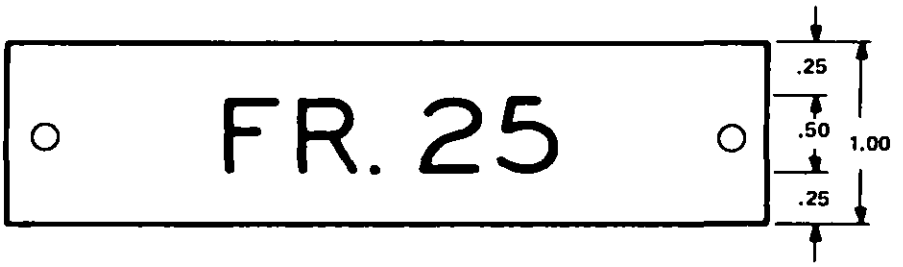


FIGURE 4. Typical builders nameplate.

X-839 B

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TYPICAL SINGLE LINE LABEL PLATE
(LIMITED TO 16-LETTER T.TLES)



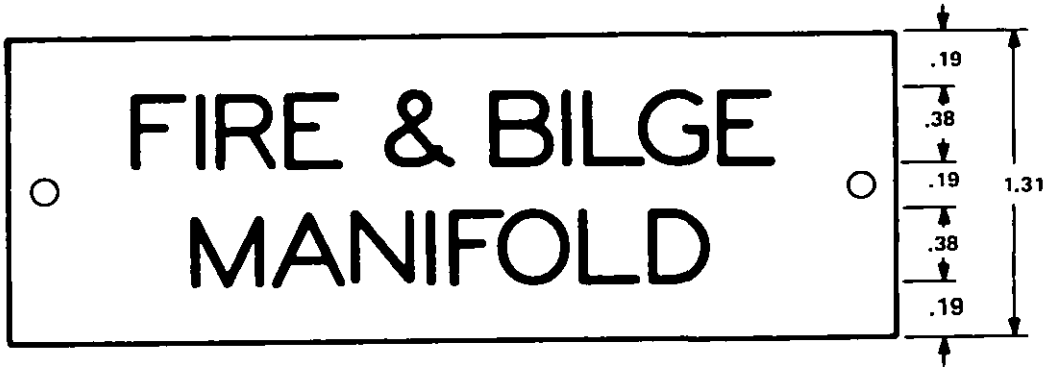
TYPICAL DOUBLE LINE LABEL PLATE
(FOR TITLES OF MORE THAN 16 LETTERS)

FIGURE 5. Typical bulkhead, frame, door, hatch, and manhole lable plates.

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TYPICAL SINGLE LINE LABEL PLATE
(LIMITED TO 16-LETTER TITLE)



TYPICAL DOUBLE LINE LABEL PLATE
(FOR TITLES OF MORE THAN 16 LETTERS)

FIGURE 6. Typical piping fittings labels.

X-841 B

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-11745D	Marking and Labeling of U.S. Army Marine Craft

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph No. and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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